

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester MCA (CBCS) Degree Examinations
 March 2023

22MCA102 - ADVANCED DATABASE SYSTEMS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from each Unit*

Unit – I		Marks	BT*	CO*	PO*										
1. a)	What are Data Models? What is the purpose of using them? Explain the different types of Data Models.	8	L2	1	1										
b)	Write short notes on the following: i) Entity sets and Relationship sets, ii) Attributes and Keys.	8	L2	1	1										
c)	Match the elements of Set A with that of Set B														
	<table><tr><th>Set A</th><th>Set B</th></tr><tr><td>Total Participation</td><td>Internal Schema</td></tr><tr><td>Logic Design</td><td>Data Model Mapping</td></tr><tr><td>Meta Data</td><td>Existence Dependency</td></tr><tr><td>Physical Design</td><td>DBMS Catalog</td></tr></table>	Set A	Set B	Total Participation	Internal Schema	Logic Design	Data Model Mapping	Meta Data	Existence Dependency	Physical Design	DBMS Catalog	4	L1	1	1
Set A	Set B														
Total Participation	Internal Schema														
Logic Design	Data Model Mapping														
Meta Data	Existence Dependency														
Physical Design	DBMS Catalog														
2. a)	What are the implications of using DBMS approach? Explain.	8	L2	1	1										
b)	Explain the following with respect to ER model. i) Role Names, ii) Cardinality	8	L2	1	1										
c)	Give one word for the following statements according to DBMS terminologies: (i) A data file in which all the records have the same record format, that is, the same field names, field lengths, and data types. (ii) The values appearing in given attributes of any tuple in the referencing relation must likewise occur in specified attributes of at least one tuple in the referenced relation. (iii) The logical design, and the snapshot of the data at a given instant in time. (iv) A collection of interrelated records.	4	L1	1	1										
Unit – II															
3. a)	With suitable examples, show how can the following be achieved in SQL. i) Setting default values, ii) Using CHECK constraint, iii) Renaming columns, iv) Updating the record/s in a table.	6	L2	2	1										
b)	What are Joins? Explain the advantages of using Joins in SQL.	4	L2	2	1										
c)	Consider the following tables and write the SQL for the following: PROJECT (Pnumber, Pname, Plocation, Dnumber) EMPLOYEE (SSN, Name, Bdate, Addr, Sex, Salary, SuperSSN, Dno) WORKS_ON(Essn, Pnno, Hours) DEPENDENT (Essn, Dependent_name, Sex, Bdata, Relationship) i) Retrieve the name and address of all employees who work for the 'Research Department'. ii) For each employee, retrieve the employee's first and last name and first and last name of his or her immediate supervisor. iii) Retrieve list of employees and the projects they are working on, ordered by department and within each department, ordered alphabetically by last name, first name. iv) Retrieve all employees whose address is in Houston, Texas. v) Find the sum of the salaries of all employees, the maximum salary, the minimum salary and the average salary.	10	L2	2	1										
4. a)	Give a brief note on the following Aggregate functions in SQL. i) MIN, ii) MAX, iii) COUNT, iv) SUM, v) AVG.	10	L1	2	1										

P.T.O.

- b) What are correlated subqueries? How are they different from normal subqueries? Illustrate 5 L2 2 1
- c) Explain the different clauses of SELECT – FROM – WHERE – GROUP – HAVING with an example for each. 5 L2 2 1

Unit – III

5. a) What is stored procedure? Write the syntax for creating stored procedure. 5 L2 3 2
- b) Write a program in PL/SQL to retrieve the records from the employees table and display them using cursors. Employees table has Id, Firstname and Salary. 10 L2 3 2
- c) Explain the following wrt PL/SQL:
i) SQL%NOTFOUND ii) %ROWTYPE 5 L2 3 1
6. a) What is a Cursor? Explain the different types of Cursors with examples. 5 L2 3 1
- b) How user defined exceptions can be implemented in PL/SQL? Explain. 5 L1 3 1
- c) What are Triggers? What are the benefits of Triggers? Give the complete syntax of creating a Trigger. 10 L2 3 1

Unit – IV

- 7 a) Explain the ACID properties of a Transaction. 5 L1 4 1
- b) List the different informal design guidelines for Relational Schemas 5 L1 4 1
- c) What is the need for normalization? Explain the first, second and third normal forms with examples. 10 L2 4 1
- 8 a) What are the different states of Transaction? Explain with a transition diagram. 5 L1 4 1
- b) Write short notes on Functional Dependency. 5 L1 4 1
- c) Convert the following unnormalized table into 1NF, 2NF and 3NF.

EMP_ID	EMP_NAME	EMP_PHONE	EMP_STATE
14	John	7272826385, 9064738238	UP
20	Harry	8574783832	Bihar
12	Sam	7390372389, 8589830302	Punjab

10 L2 4 1

Unit – V

9. a) What are NoSQL databases? What are their characteristic features? Briefly explain the different types of NoSQL databases? 10 L1 5 1
- b) Write MongoDB commands for the following:
i) Insert a document into a collection called "Movies" with the properties namely Id, Name, Category, Rating (out of 10),
ii) Write a query to return the 3 highest rated movies,
iii) Write a query that returns the details of those movies which has a rating of more than 5 and belongs to category "Animation",
iv) Replace the contents of entire record which has object id 2 using Save method,
v) Drop collections called movies. 10 L2 5 1
- 10 a) What are the advantages of NoSQL over traditional RDBMS? Explain. 10 L1 5 1
- b) Explain the following MongoDB commands with purpose, syntax and example: i) Insert, ii) Save, iii) Update, iv) Remove, v) Find. 10 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
